# Heet Mehta

Machine Learning Developer | Full-Stack Engineer

Anand, Gujarat, India | +91 93164 28942 | heetmehta.work@gmail.com
Portfolio | GitHub | LinkedIn | LeetCode

## SUMMARY

Third-year B.Tech Computer Science student at CHARUSAT University (CGPA 7.86) building and evaluating LLM systems. Fine-tuned small language models with QLoRA, designed abstention-based trust evaluation for financial tasks, and shipped distributed full-stack systems. Open-source contributor to Ollama and AirLLM. Seeking a Machine Learning Developer internship.

## EDUCATION

**B.Tech, Computer Science and Engineering** – CHARUSAT University (CSPIT), Anand, Gujarat 2024 – 2028
CGPA: 7.86 / 10 | Coursework: Probability & Statistics, Data Structures & Algorithms, DBMS, Computer Networks, Python for Data Science
HSC – Science, GSEB – 95.40 Percentile (2024) | SSC, GSEB – 98.45 Percentile (2022)

## TECHNICAL SKILLS

**Machine Learning / AI:** Hugging Face, QLoRA, LLM Fine-tuning, TensorFlow, scikit-learn, Deep Learning, NumPy, Pandas, Feature Engineering, LLM APIs, Prompt Engineering
**Evaluation:** F1, ROC-AUC, Precision-Recall, Confusion Matrix, McNemar Test, Bootstrap Confidence Intervals
**Languages:** Python, Go, TypeScript, JavaScript, C, C++, Java
**Web & Backend:** React, Next.js, Node.js, Express.js, REST APIs, WebSocket
**Databases:** PostgreSQL, MongoDB, Redis, PgBouncer
**DevOps & Tools:** Docker, Nginx, GitHub Actions, CI/CD, Git, Linux

## EXPERIENCE

**Machine Learning Intern** – CHARUSAT University (CSPIT), Anand, Gujarat Jan 2026 – Apr 2026

- Trained and evaluated supervised ML models (Random Forest, SVM, Logistic Regression) on structured research datasets; improved F1 score by 12% through hyperparameter tuning and 5-fold cross-validation.
- Built end-to-end feature engineering pipelines (null imputation, scaling, one-hot encoding), reducing data preparation time by about 40%; produced ROC-AUC, precision-recall and confusion matrix reports for faculty.

## PROJECTS

**TrustFin-SLM** – Trustworthy Small Language Models for Financial Tasks Jul 2026 – Present
*Python, Hugging Face, Qwen3-1.7B, QLoRA* | *GitHub*

- Fine-tuned Qwen3-1.7B with balanced joint QLoRA across 16 baseline experiments, selecting the backbone on validation data only (FinQA execution score 0.675, FiQA $R^2$ 0.730).
- Built a reproducible 4-benchmark evaluation pipeline (FinQA, FiQA, FOMC, FinTrust) with SHA-256 frozen data contracts and 68 validation gates, keeping protected test sets out of training and threshold tuning.
- Proposed the EVG-CAS abstention method: a frozen threshold cut protected-test answered-error by 31.0% at 80% coverage, validated with paired McNemar tests and bootstrap confidence intervals; research paper draft in progress.

**Code Pie** (formerly BinaryBattles) – Competitive Programming Platform Mar 2026 – Present
*Node.js, React, TypeScript, PostgreSQL, Redis, Bull, Docker, Nginx, PgBouncer* | *GitHub*

- Contributor in an 18-member team; owned the contest module and its architecture, including submission-queue capacity and rate thresholds across Redis/Bull queues and isolated Docker workers.
- Scaled to a distributed 3-node setup (Nginx load balancer, PgBouncer, Redis workers) sustaining 1,000+ requests per second across submission queue stages (worker runner, Redis queue/cache, PostgreSQL) and contest-page APIs.
- Built the contest workspace on Monaco Editor with C/C++, Python and SQL judging; in-progress code persists in localStorage.

**AutoDevs** – Open Source Developer CLI (featured on Product Hunt) 2025 – Present
*Go, Node.js, TypeScript, GitHub API, Docker* | *Website* | *GitHub*

- Built a cross-platform developer-environment bootstrapper in Go that detects a project's stack and installs missing runtimes, dependencies and tools in one command (40+ packages, 9 developer-role profiles, 3 OS targets).
- Published on npm, Homebrew, Scoop and Docker; GitHub Scanner scores language/framework confidence from public repositories using frequency-weighted heuristics and outputs a skill matrix with upgrade paths.

**Coastal Guardian** – Deep Learning for Coastal Threat Classification 2025
*Python, TensorFlow, CNN, ResNet50, scikit-learn* | *GitHub*

- Designed a CNN on NASA satellite imagery to classify 4 coastal threat categories (erosion, pollution, algal bloom, clear) using ResNet50 transfer learning and augmentation on imbalanced classes.
- Achieved 87%+ validation accuracy on 2,400 image patches, evaluated with confusion matrix, ROC-AUC and precision-recall.

## OPEN SOURCE CONTRIBUTIONS

- **AirLLM** – PR #339 (GitHub): fixed layer-count parsing for nested checkpoint prefixes (VLM/multimodal models), removing a ValueError crash when splitting checkpoints; added unit tests across supported architectures.
- **Ollama** – PR #14992 (GitHub), Mar 2026: removed duplicate TopLogprobs validation dead code across GenerateHandler and ChatHandler in a Go LLM inference engine.

## LEADERSHIP & CERTIFICATIONS

- **Developer and ML Tutor**, Open Source Club, CHARUSAT University (Website) – built the club website; teach ML to members.
- **Coordinator**, C Titans Hackathon (50+ participants) and SUPERNOVA coding event, CHARUSAT.
- Analytics with Python – Coursera (2026); Algorithm Design – Coursera (2026); Foundations of Data Structures and Algorithm Analysis – Packt (2024).